

Postural versus chair design impacts upon interface pressure

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Abstract

An investigation of postural and chair design impacts upon seat pan interface pressure has been performed in an effort to identify whether differences in posture or chair design result in greater pressure differences. Investigation of postural variables focused on trunk–thigh angle and use of armrests. Twelve ergonomic office chairs were used to assess chair design differences. Both male and female subjects were included. Gender effects were controlled through use of a repeated Latin square design, with squares defined by gender. Significant gender-based interaction was observed amongst postural treatments and chair effects. Postural treatments, chairs designs, and participant effects all resulted in significant interface pressure differences, though gender-based interaction yielded some non-additivity of results between males and females. The final conclusion drawn from the results is that chair design differences had the greatest effect on seat pan interface pressure, followed by participant effects, and lastly postural treatments.

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1. Introduction

1.1. Seated posture—a biomechanical and physiological description

Published estimates have indicated that almost 75% of work in industrial countries is performed while seated, a proportion which strongly suggests a certain degree of importance in studying the science of sitting (Treaster and Marras, 1987). When a seated posture is assumed, the majority of the body's weight is placed upon the supporting area of the ischial tuberosities of the pelvis and the tissues in their proximity (Schoberth, 1962; Chaffin et al., 1999). As a person sits, the pelvis rotates backwards, the lumbar spine may flatten, and the ischial tuberosities become the main weight-bearing structure in close contact with the

seating surface (Andersson et al., 1979; Congleton et al., 1988; Chaffin et al., 1999; Sember, 1994). The tissue of the gluteus maximus muscles as well as local deposits of adipose tissue form a cushioning layer around the posterior of the pelvic structure and beneath the ischial tuberosities. This layer is often quite thick while standing, though when a person assumes a seated posture the ischial tuberosities begin to bear the weight of the upper body and compress the surrounding soft tissues until a relatively thin layer remains to provide cushioning and support (Sember, 1994). As pressure under the ischial tuberosities increases, blood flow to tissues of the region may be restricted as tissue compression exceeds hydrostatic capillary pressure, an effect which may manifest symptomatically as a sensory indication of pain or discomfort beyond a certain threshold (Sember, 1994; Yarkony, 1994; Carison et al., 1995). Variables affecting tissue compression and interface pressure in the seat pan may include personal factors (e.g. anthropometric variables), postural factors, as well as chair design factors.

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1.2. Personal factors

Personal factors represented as variables of anthropometry may have a significant impact on human–chair interaction. Those that have been reported upon extensively with regard to seat pan interface pressure include subject gender and measures of body build or composition. Gender-based differences have been reported with males having experienced greater interface pressures than females (Yang et al., 1984; Gyi and Porter, 1999). Subject stature has also been associated with interface pressure, with increased statures related with increased pressures (Yang et al., 1984). Differences in interface pressure distributions have also been associated with variation in body composition, as quantified using a variety of indices such as the Reciprocal Ponder Index (RPI) (a value relating body mass and length), the Body Mass Index (BMI), and general categorizations of thin, average, and obese (Garber and Krouskop, 1982; Gyi and Porter, 1999).

1.3. Postural factors

Past research efforts regarding seat pan interface pressure have generally focused upon posture as an outcome variable or have utilized single constant postures, indicating an opportunity for further research of posture as a controlled independent variable. However, posture has been examined as a controlled variable with regard to other outcomes, the results of which suggest potential postural variables of interest to this study. Postural variables have been reported to affect internal physiological conditions, subjective ratings of comfort or discomfort, and secondary effects such as metrics of productivity (Andersson and Örtengren, 1974a,b; Andersson et al., 1974a,b; Bhatnager et al., 1985; Chaffin et al., 1999). Postures assumed in the workplace may include forward leaning postures, upright postures, as well as reclined postures (Mandal, 1981; Chaffin et al., 1999). From a biomechanical and physiological perspective, studies have shown posture changes associated with variation in the backrest angle of a chair can have a significant effect on the body. Increased (reclined) backrest angles (e.g. angles ranging from 100° to 120° from the horizontal) have been associated with reduced spinal disc pressure (Andersson and Örtengren, 1974a,b; Andersson et al., 1974a,b; Chaffin et al., 1999). Disc pressures at 120° were the lowest, being only 50% of those observed at 90°, indicating that increased backrest angle could be beneficial at reducing disc pressure in seated populations. In addition, the use of armrests has been shown to have an effect in the reduction of disc pressures (Andersson et al., 1974a). Increased backrest angle has also been associated with reduced muscle activity in the back muscles when measured by electromyography (EMG) (Knutsson et al., 1966; Rosemeyer, 1971; Chaffin et al., 1999).

1.4. Chair design factors

To combat potentially negative postural, biomechanical and physiological changes due to seated posture, numerous variations in chair design have been used. Potential design variables are numerous, and have included variations in seat cushioning, seat fabrics, seat pan designs, backrest designs, back rest and seat pan adjustability angles, lumbar support, and seat height to name just a few. Some of these design variables been shown to have quantifiable impacts upon seat pan interface pressure. Specifically, many studies have indicated significant differences between varying degrees of cushion thickness, density and composition, as well as chair contouring (Garber and Krouskop, 1982; Yang et al., 1984; Congleton et al., 1988; Sember, 1994; Gyi and Porter, 1999).

1.5. Interface pressure measurement

Historically, measures of buttock thigh compression have been used as a metric either assumed to be associated with seated discomfort or directly correlated with it based upon experimental subjective ratings of comfort or discomfort (Congleton et al., 1988; Ebe and Griffin, 2001; Gyi and Porter, 1999; Porter et al., 2003). Though past research has indicated a possible relationship between pressure and discomfort, recent studies have suggested that any such relationship is neither simple nor direct, and may exist only for certain population subsets (Gyi and Porter, 1999; Porter et al., 2003). Potentially confounding the issue is that seated discomfort appears to be strongly associated with fatigue (Helander and Zhang, 1997). However, the quantitative and objective collection of interface pressure data have been identified and corroborated repeatedly as an appropriate metric for assessing the impact of seating related variables such as posture, seat construction and structural support of the body, which from an objective physiological standpoint have a direct impact on buttock–thigh tissue compression (Sember, 1994; Yarkony, 1994; Carison et al., 1995; Gyi and Porter, 1999; Porter et al., 2003). Therefore, interface pressure measurement continues to be a useful tool in assessing tissue compression due to factors affecting seat pan interface pressure.

1.6. Specific aims and rationale

The specific aims of this study were to further investigate the impacts of personal, postural, and design factors upon seat pan interface pressure. Personal factors of interest included measures of body composition or build. Postural factors of interest included trunk–thigh angle and the use of arm rests. Chair design factors were investigated in a broad sense using actual chairs used in present-day office settings (rather than use of a laboratory apparatus). Personal factors were included in the investigation as past research suggested they may account for a significant and measurable portion of experimental variability. Postural

factors were studied as they are alterable conditions of seated work, and could provide results that can be leveraged in the real world to reduce tissue compression. Chair design factors were investigated using real world examples as they represented seating devices in current use at the time of the study and would be an alterable conditions of the existing workplace. It was of particular interest to determine if chair design differences or postural factors account for greater differences in seat pan interface pressure.

2. Materials and methods

2.1. Materials

Equipment used in the course of this study included an interface pressure mapping system, a Hitachi M-133T/1000 notebook computer, four digital weight scales, a standard medical weight scale, a digital electro-goniometer, a carpenter's bubble level, and 12 ergonomic office chairs.

The interface pressure mapping system used in the study was manufactured by the XsensorTM Technology Corporation. The system implemented a thin profile interface mat constructed using a capacitive elastomer sensor technology, with each pressure sensor consisting of a dielectric between two conductive elements. The mat consisted of a matrix of 1296 pressure sensors, mounted within a thin vinyl mat. The sensor mat's accuracy was rated at ± 10 mmHg for observed pressure measurements and inter-trial comparisons. With regard to hysteresis (the retardation of sensor accuracy due to compression and subsequent decompression of the device), the system was designed with integral software correction technology, implemented within the system's software interface through the use of a calibration file which was generated when the device was returned to the manufacturer for calibration and certification prior to use in the experiment. The calibration range used by the manufacturer was 10–200 mmHg. The pressure data sampled by the mat could be observed in real-time, with a refresh rate of 5000 sensor samples per second (or 3.9 pad samples per second, for the 1296 sensor mat).

The ergonomic office chairs selected for inclusion in the study were sourced from various international manufacturers. All of the chairs included were in widespread distribution and use at the time of this study. Each chair was chosen for its differences from the others, whether it was as dramatic a difference as its fundamental engineering design or as simple a difference as a variation in the type or thickness of the seat pans' foam or fabric. In an effort to avoid commercialism, no mention will be made of the chair makes or models. Instead, a brief description of their basic designs is tabulated with a corresponding "chair code" which is used for chair identification. The chair descriptions themselves are purposely kept brief and general since with a sufficient description a knowledgeable ergonomist could easily recognize many of the chairs. This type of identification is deemed to be appropriate since the goal of

this study was not a detailed investigation of chair design variables, but rather a comparison of whether basic differences in chair designs or changes in subject posture had a greater impact on subject-seat interface pressure distributions. Table 1 is a list of the chair descriptions and the corresponding identifier codes used in the experiment. All chairs had a five-point base, a hydraulic/pneumatic main support cylinder, a backrest, and armrests.

2.2. Participants

Twenty-four participants took part in this experiment, including 12 males and 12 females. The participants included both members of the local university student body as well as members of the general population. Anthropometric data collected from the participants included stature and mass, from which BMI and RPI values were calculated. Statistics regarding these variables are reported in Table 2.

2.3. Experimental design

A Repeated Latin Square (RLS) experimental design was implemented for this experiment, using a factorial treatment structure (Lenter and Bishop, 1993). Two matching Latin squares were created, one for each gender, with testing of males and females performed independently within their own squares. This design permitted controlled investigation of potential gender effects, including gender-based interactions with experimental treatments. Should gender interactions be found to exist, each gender could be analyzed separately. The Latin squares were organized based upon the use of 12 subjects for each gender, 12 chairs, and six postural treatments. The randomized Latin squares used in this experiment were generated based on the process and the random number table detailed in Lenter and Bishop (1993). This resulted in 144 data points for each gender (12 subjects $\times$ 12 chairs), and 288 data points total.

The six postural treatments were created using a combination of three different trunk–thigh angles (100°, 110°, 120°) with each angle evaluated both with and without the use of armrests, thus forming a 2 $\times$ 3 factorial treatment structure (Lenter and Bishop, 1993). These specific angles and the use of armrests were of particular interest due to their use in prior studies which reported significant findings for other physiological variables associated with increased (reclined) postural angle (Andersson and Örtengren, 1974a,b; Andersson et al., 1974a,b; Chaffin et al., 1999). With regard to seat pan interface pressure, the interest in potential pressure differences due to postural angle was based upon off-loading of body weight onto the backrest and armrests. As the backrest reclines to a greater degree, more of the upper body weight is borne by the backrest support, reducing the percentage of body weight transferred downwards to the seat pan. Seat pan interface pressure occurs due to the

Table 1
Table of chair codes and basic descriptions

Chair code	Seat pan foam depth (cm)	Foam type	Fabric type	Seat pan contouring	Backrest features	Armrest adjustability
C1	5.1	Traditional foam	Knitted fabric	Slight contouring	Moderate contouring, supported shoulders	Adjustable armrest height and angle
C2	5.1	Traditional foam	Woven fabric	Slight contouring	Medium height, only slight shoulder support	Adjustable armrest height and angle
C3	N/A	Tensile mesh	Tensile mesh	Slight contouring	Supported shoulders, adj. lumbar height	Adjustable armrest height and angle
C4	3.8	Traditional foam	Woven fabric	Slight contouring	Supported shoulders	Adjustable armrest height and angle
C5	6.3	Traditional foam	Knitted fabric	Highly contoured	Supported shoulders	Adjustable armrest height and angle
C6	5.0	Traditional foam	Woven fabric	Slight contouring	Supported shoulders	Non-adjustable
C7	5.1	Traditional foam	Woven fabric	Slight contouring	Moderate contouring, supported shoulders	Adjustable armrest height and angle
C8	3.8	Traditional foam	Woven fabric	Medium contouring	Flexible backrest design, supported shoulders	Adjustable armrest height and angle
C9	5.1	Traditional foam	Knitted fabric	Slight contouring	Medium height, supported shoulders	Adjustable armrest height and angle
C10	3.8	Traditional foam	Woven fabric	Slight contouring	Medium height, supported shoulders	Adjustable armrest height
C11	4.4	Traditional foam	Woven fabric	Medium contouring	Short height, no shoulder support	Adjustable armrest height and angle
C12	6.3	Visco-elastic foam	Knitted fabric	Highly contoured	Supported shoulders	Adjustable armrest height and angle

Table 2
Table of participant information

Gender	Variable	N	Mean	Minimum	Maximum	Std. dev.	Skewness
Male	Stature (cm)	12	177.44	167.64	183.64	5.17	−0.77
	Mass (kg)	12	86.63	74.50	124.28	14.25	1.90
	BMI	12	27.51	24.03	40.46	4.43	2.59
	RPI	12	40.27	35.12	42.00	1.87	−2.04
Female	Stature (cm)	12	163.88	158.12	170.18	3.35	0.16
	Mass (kg)	12	63.14	51.71	72.80	7.77	−0.15
	BMI	12	23.48	19.92	27.29	2.60	0.30
	RPI	12	41.27	39.05	43.77	1.48	−0.19

BMI: Body Mass Index = (mass (kg)/stature (cm)²) × (10,000). RPI: Reciprocal Ponder Index = stature (cm)/mass (kg)^{1/3}.

downward force of gravity upon the body. This results in the upper body weight being transferred via the spine to the pelvic structure. The pelvis in turn bears the resultant downward force of the upper body, with peak seat pan interface pressures occurring under the ischial tuberosities (Andersson et al., 1979; Congleton et al., 1988; Chaffin et al., 1999; Sember, 1994). Since increased offloading of upper body weight due to backrest angle and armrest use could reduce the resultant downward force borne by the pelvic structure, increased trunk–thigh angle and armrest usage was expected to demonstrate a reduction in seat pan interface pressures.

The primary dependent variables measured with regard to interface pressure distributions included peak and mean pressure. Peak pressure was defined as the highest point of

pressure observed in the interface pressure distribution during the testing period. This typically occurs under the location of the ischial tuberosities. The resolution of the sensor mat used was 1.27 cm², thus this would be the highest 1.27 cm² cell in the 45.7 cm × 45.7 cm (2088.5 cm²) mat. Mean pressure was defined as the averaged pressure observed across all activated cells in the sensor mat (note that only cells receiving 5 mmHg of pressure or more were considered “activated”).

2.4. Testing methodology

The testing methodology used in the experiment was approved for the use of human participants in research by the Texas A&M University Institutional Review Board.

The experimental procedure included initial preparation of the testing equipment, setup of the test participants, and collection of interface pressure distributions. During each data collection session, the seat pan interface pressures were recorded digitally, via the sensing device's computer interface and software. There was one session per participant. Each session was conducted using the following protocol:

1. Each participant was taken through the informed consent process.
2. If the participant wore clothing with thick material, seams or rivets (i.e. blue jeans), they were asked to don a pair of seamless cotton exercise pants.
3. Basic demographic and anthropometric variables such as gender, stature and mass were collected.
4. Scales used to measure the mass present under each arm and each foot were activated and zeroed.
5. Two scales were positioned to measure the mass of the participant supported by the feet.
6. If the postural treatment prescribed the use of armrests, a scale was placed on each armrest, and the armrests were adjusted such that the surface of the scales were at the subject's seated elbow height. These were used to measure the weight of the subject off-loaded to the armrests, and to ensure that the subject was not leaning to either side.
7. The sensor mat was then placed into the chair.
8. The participant was requested to sit into the seat carefully, being careful not to wrinkle or disrupt the placing of the sensor mat.
9. The participant then placed his/her feet onto the foot scales. If using an armrest treatment, they were also asked to place their arms upon the armrest scales.
10. Standard anatomical landmarks were used to position each participant's trunk–thigh angle into the three treatment angles.
11. Once the subject's posture and the chair were both properly adjusted, and the subject had been seated for 10 min (to allow the foam of the seat pan to compress due to the weight of the subject) the subject–seat interface pressure distribution was recorded using the Xsensor software system. One hundred frames of pressure data were collected, at a sampling rate of 10 Hz.
12. This process was repeated for each of the 12 chairs and the specified treatments, as designated by the Latin square treatment design (Fig. 1).

The process of positioning each subject into each of the postural treatments involved adjustment of the chairs' seat pan angles, seat pan heights and trunk–thigh angles. The anatomical landmarks used for trunk–thigh angle included the acromion, greater trochanter of the femur and the lateral condyle of the femur. The greater trochanter of the femur and the lateral condyle of the femur were used as landmarks with a bubble meter to constrain the subject's



Fig. 1.

thigh to a horizontal level. The thigh was maintained at a horizontal level so that outcomes would have a common reference with a horizontal baseline due to the experimental focus on examination of trunk–thigh and armrest postural variables rather than seat pan tilt. In addition, the lateral condyle and the lateral malleolus were used as landmarks along with a bubble level to ensure that the lower leg was constrained to a vertical. This was done to ensure that each participant would have a knee angle of 90°, controlling for lower body posture, which was shown to have significant effects upon peak and mean pressures during preliminary testing. The postural setup process was repeated for both right and left sides of the body.

2.5. Statistical analysis

Statistical analyses were performed using Statistica 6.0 for Windows. Analyses performed included principal components factor analysis, general linear model analysis of variance (GLM ANOVA), post hoc testing with Tukey's Honest Significant Difference (Tukey's HSD), and correlation assessment. For all pertinent tests significance was determined using a 0.05 level.

3. Results

3.1. Factor analysis

There were two primary dependent variables in this study including mean and peak pressure observed in the seat pan. As both of the primary dependent variables in

this study were outcomes related to interface pressure distribution, a preliminary factor analysis was performed to determine if the outcomes could be explained and presented as a single factor for evaluating the impacts upon interface pressure due to each of the main effects under investigation. A principal components factor analysis was performed using an eigenvalue criterion of 1.0 for factor decisions, and unrotated factor loadings. Results indicated mean and peak pressure could indeed be represented as a single “Pressure Factor,” with factor loadings of 0.87 for each of the two variables, an eigenvalue of 1.52, and explanation of 76.2% of the total variance.

3.2. Analysis of variance

A mixed model ANOVA was conducted using the pressure factor variable created through the principal components factor analysis. The ANOVA model used was based on the RLS experimental design and the factorial treatment structure, and was created using the protocol for RLS ANOVA and factorial treatment structures in block-based designs as set forth by Lenter and Bishop (1993). Gender defined the two squares, participants (included as a random factor) were nested within gender while both the chair factor (also random)

and the factorial treatment structure were crossed with the square effect since the same chairs and treatments were used for both squares. Table 3 presents the RLS ANOVA model generated, as well as the results of the analysis. The Satterthwaite (1946) method was used for testing of random effects in the mixed model design.

The RLS ANOVA allowed for testing of the gender factor to determine if there were gender-based interactions with either the treatment structure or the chair factor. Results indicated that significant interaction existed between the square effect (gender) and both chairs and treatments, thus additional independent Latin square analyses were done for each gender, also presented in Table 3. Results of the independent gender assessments indicated significant differences within the male square for all effects except the armrest $\times$ angle interaction term. Results for the female square were similar, with an additional lack of significance for the armrest effect.

The use of a mixed model ANOVA also allowed for a variance component analysis of the portions of random variance associated with each of the random factors and the error term. The chair factor accounted for 44% of the random variance, participant (nested within gender) for 25%, the gender by chair interaction term for 4%, while the error term accounted for 28% of the random variance.

Table 3
Mixed model ANOVA results for repeated Latin square with factorial treatments

Analysis	Source ^b	Effect ^b	SS ^b	df ^b	MS ^b	DSE df ^c	DSE MS ^d	F	p
Combined gender RLS ^a ($r^2 = 0.83$) ^b	Gender	Fixed	55.406	1	55.406	25.789	2.840	19.512	0.000
	Participant (gender)	Random	55.067	22	2.503	232.000	0.215	11.633	0.000
	Chair	Random	94.618	11	8.602	11.000	0.552	15.593	0.000
	Armrest	Fixed	0.541	1	0.541	232.000	0.215	2.516	0.114
	Angle	Fixed	21.857	2	10.928	232.000	0.215	50.788	0.000
	Armrest $\times$ Angle	Fixed	0.352	2	0.176	232.000	0.215	0.818	0.443
	Gender $\times$ Chair	Random	6.068	11	0.552	232.000	0.215	2.564	0.004
	Gender $\times$ Armrest	Fixed	1.304	1	1.304	232.000	0.215	6.061	0.015
	Gender $\times$ Angle	Fixed	1.775	2	0.887	232.000	0.215	4.124	0.017
	Gender $\times$ Armrest $\times$ Angle	Fixed	0.091	2	0.046	232.000	0.215	0.212	0.809
	Error		49.921	232	0.215				
Male square ($r^2 = 0.78$) ^b	Chair	Random	45.228	11	4.112	116.0	0.166	24.838	0.000
	Participant	Random	16.416	11	1.492	116.0	0.166	9.015	0.000
	Angle	Fixed	5.837	2	2.919	116.0	0.166	17.631	0.000
	Armrest	Fixed	1.763	1	1.763	116.0	0.166	10.651	0.001
	Armrest $\times$ Angle	Fixed	0.305	2	0.153	116.0	0.166	0.922	0.401
	Error		19.203	116	0.166				
Female square ($r^2 = 0.78$) ^b	Chair	Random	55.457	11	5.042	116.0	0.265	19.038	0.000
	Participant	Random	38.652	11	3.514	116.0	0.265	13.269	0.000
	Angle	Fixed	17.794	2	8.897	116.0	0.265	33.598	0.000
	Armrest	Fixed	0.082	1	0.082	116.0	0.265	0.312	0.578
	Armrest $\times$ Angle	Fixed	0.138	2	0.069	116.0	0.265	0.260	0.771
	Error		30.718	116	0.265				

Both the repeated Latin square (combined) analysis and the independent square analyses are provided (where gender represents the two squares).

^aRepeated Latin Square. The RLS model allows combination of separate Latin squares for examination of square (gender) effects and interactions.

^bCoefficient of determination (r^2); Source of Variation (Source); Effect Model (Effect); Sum of Squares (SS); Degrees of Freedom (df); Mean Squares (MS).

^cDenominator synthesis error degrees of freedom (DSE df). Used by Satterthwaite (1946) mixed model ANOVA for testing random factor significance.

^dDenominator synthesis error mean square (DSE MS). Used by Satterthwaite (1946) mixed model ANOVA for testing random factor significance.

3.3. Significant differences and post hoc testing

Significant differences identified by the ANOVA were further investigated through post hoc tests using Tukey's HSD. Differences in the square effect (gender) were identified from the RLS ANOVA, while all other factors were tested based upon independent square Tukey's analyses due to gender-based interactions.

The significant gender-based square effect was examined, revealing that females experienced lower pressure factor values than males. There were no significant differences due to armrest usage for females, though males did experience a rather slight but statistically significant difference with armrest usage yielding some reduction in pressure factor values. Increased (reclined) trunk–thigh angle was found to significantly reduce pressure factor values for both males and females (tested separately), as demonstrated in Fig. 2, which plots pressure factor means by trunk–thigh angle separately for each gender, and identifies Tukey's HSD groupings by letter (means sharing the same letter are not significantly different, while those with different letters are). The lines plotted for each series (gender) in Fig. 2 facilitate identification of gender-based interaction (non-additivity). Significant differences were also observed amongst the chairs (tested separately for each gender). The chairs were also plotted by pressure factor means separately for each gender as depicted in Fig. 3. Fig. 3 is also plotted with lines for each gender series to assist with determination of where non-additivity has occurred, which was most apparent for chairs 10, 3 and 2. As with Fig. 2, Tukey's HSD defined homogenous

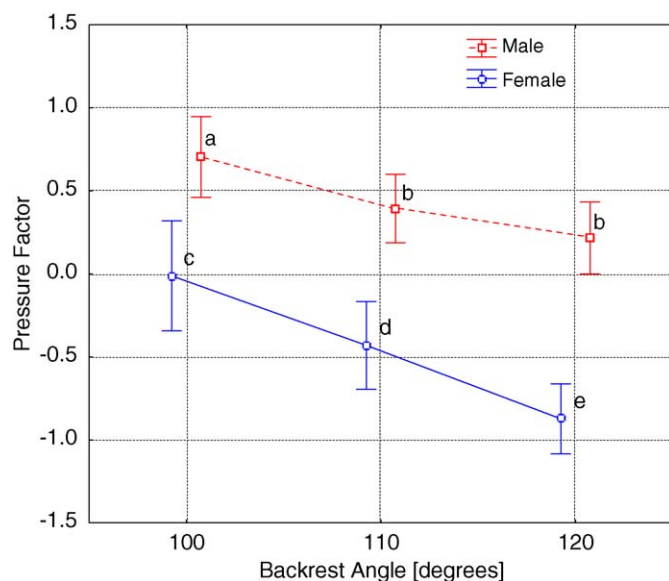


Fig. 2. Pressure factor means and independent Tukey's HSD defined homogenous subgroups for each gender by postural angle. Means grouped by the same letters were not significantly different, while those with different letters were significantly different. Analyses were conducted separately for each gender. Lines are present to illustrate presence or lack of additivity due to gender-based interactions.

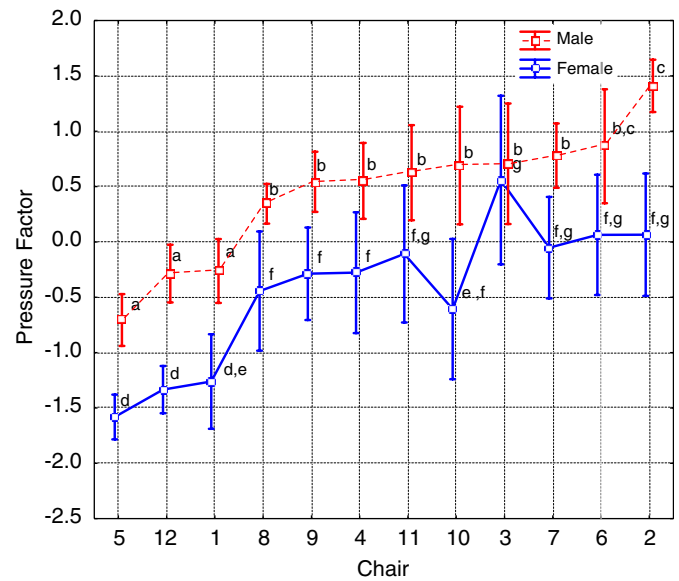


Fig. 3. Pressure factor means and independent Tukey's HSD defined homogenous subgroups for each gender by chair. Means grouped by the same letters were not significantly different, while those with different letters were significantly different. Analyses were conducted separately for each gender. Lines are present to illustrate presence or lack of additivity due to gender-based interactions.

subgroups are identified by letter, with means sharing identical letters lacking significant difference, while those with different letters were identified as significantly different.

Following post hoc testing, the magnitude of mean difference between the maximum and minimum pressure factor mean within each of the independent variables were examined by gender. The mean pressure factor difference between male and female results was 0.88. For males the chair factor appears to have resulted in the greatest difference between the lowest and highest values (difference = 2.12), followed by participant effects (difference = 1.24), angle (difference = 0.48), and finally armrest usage (difference = 0.22). Female results were similar, with chairs having the greatest impact on pressure factor values (difference = 2.14), with participant differences next (difference = 1.59), and finally angle (difference = 0.86).

3.4. Correlations

Spearman's rank order correlations were examined amongst several variables in an effort to determine if there were any relationships between dependent variables or amongst potential correlates. Raw dependent variable outcomes included significant correlation ($p < 0.05$) between mean and peak pressure outcomes ($r = 0.63$), with little to no correlation between either mean or peak pressures and active cell counts (analogous to contact surface area). The pressure factor was moderately but significantly correlated with participant mass ($r = 0.42$)

and participant stature ($r = 0.38$), with weak but significant correlations observed for BMI ($r = 0.33$) and RPI ($r = -0.21$). Moderate significant correlations were observed between active cell counts and participant BMI ($r = 0.62$), mass ($r = 0.61$), RPI ($r = 0.5$) and stature ($r = 0.48$).

4. Discussion

4.1. Discussion of data and results

The RLS ANOVA indicated that not only was there a significant difference between male and female results, but that several other factors interacted with gender. The gender differences revealed that on average male pressure factor values were higher than female values. This is possibly explained by a variety of anthropometric differences between the two genders. For example, there are differences in pelvic shape and size distributions between males and females as well as differences in the size and shape of the ischial tuberosities (NASA, 1978; Van De Graaf and Fox, 1999). Male pelvic structures are generally narrower, with ischial tuberosities closely spaced and having a tighter pubic arch, resulting in a smaller load-bearing area under the ischial tuberosities (Van De Graaf and Fox, 1999). Female pelvic structures provide a greater pubic arch and more broadly set ischial tuberosities, resulting in a wider pelvic structure which may serve to better distribute weight in the seat pan (Van De Graaf and Fox, 1999). Females also have a greater seated hip breadth and an observed lower mean mass, which also may contribute to a more effectively distributed weight (NASA, 1978). The gender-based findings of this study are in agreement with prior studies which have also found the magnitude of male pressure outcomes to be greater than those observed for females (Yang et al., 1984; Gyi and Porter, 1999).

Investigation of other gender-based interactions showed that the use of armrests did not affect female pressure distributions, and that reclined trunk–thigh angle only seemed to reduce female pressure distributions to a limited extent, with 120° showing no significant additional reduction beyond that provided at 110° . Males did experience statistically significant reduction in pressure distributions through the use of armrests though the difference was slight. Males also experienced a linear reduction in pressure factor values due to increased (reclined) backrest angle. Male pressure distributions therefore appeared to be affected by posture to a greater degree than female distributions.

Examination of gender-based differences for chairs (Table 3) revealed that for eight of out 12 chairs the gender effect was additive, with a few specific chairs resulting in significant gender by chair interaction (chairs 10, 3, 2 and to a lesser degree chair 1). Chairs were included in the RLS ANOVA as a random factor, and it is unknown

what specific properties of these four chairs may have resulted in gender-based interactions.

The observed mean pressure factor differences indicate that the chair effect resulted in the greatest difference in mean pressure factor values for both genders, with a pressure factor difference of 2.1 between the maximum and minimum chair pressure factor means for both males and females. The observed chair pressure factor difference was 4 times greater than that observed for the male angle effect, and 2.5 times greater than the female angle effect. Therefore chair design differences appear to have a greater impact on seat pan pressure interface distributions than postural angles or the use of armrests.

Though quantitative measures related to chair design were not included as variables in this study, chairs 5, 12, and 1 were observed for both genders to have the lowest mean pressure factor values. In trying to determine what design factors these three chairs may have shared, review of Table 1 indicates that all three of these chairs had seat pan foam depths between 5.1 and 6.3 cm, utilized knitted fabrics as opposed to woven fabrics or tensile mesh, had backrests offering shoulder support, and adjustable armrests. Conversely, chairs 6 and 2 resulted in the highest pressure factor values for males, while the highest female chair category was much bit boarder, including chairs 11, 3, 7, 6, and 2. Attempting to identify common qualities of these chairs from Table 1 indicates fabric type as the only consistent difference from the lowest pressures chairs. All chairs in the highest pressure grouping used either woven fabrics or tensile mesh. It is unlikely that the only chair design factor to significantly impact seat pan interface pressure is fabric type, suggesting that there are other chair design variables not recorded by this study which may have a significant effect. It is also possible that significant interactions exist amongst chair design variables, confounding identification of relationships between single design factors and pressure factor distributions.

Participant-based differences within each gender were also significant, accounting for 25% of the random variance observed in the RLS ANOVA model. Though covariates representing participant body composition were not included in the mixed model ANOVA, the moderate to weak correlations observed for subject mass, stature, and BMI with pressure factor values indicates that some quality of body composition might have had an impact on observed pressure distributions. However, there are other potential covariates unaccounted for by this study (e.g. anthropometric characteristics of the buttock–thigh region) which might also play a significant role in participant effects.

4.2. Strengths and weaknesses of the study

Potential strengths of this study included use of appropriate experimental design and statistical modeling, proper statistical methods for detecting mean differences

amongst main effect factors within the statistical model, and the inclusion of both male and female subjects.

A potential weakness of the study included the use of a posture variable treatment with a trunk–thigh angle of 120°, which may have resulted in increased sacral pressures for some subjects. A potential result of sacral loading could have been that as peak pressure was off-loaded from the ischial tuberosities it was to some degree transferred to the sacral region, mitigating the net change in peak pressure due to the increased trunk–thigh angle. This particular angle was chosen at the onset of the study due to it being the greatest angle that the included chairs would allow subjects to assume. For future research similar to this study, it is advised that researchers avoid incorporating such a large trunk–thigh angle, and find an appropriate angle less than 120° but greater than 110°, where sacral loading might not be a potential problem.

As stated in the methodology section, the current study design controlled for gender, however, it did not control or measure other potential random and/or confounding factors (e.g. various anthropometric values or population demographics). Additional information could have been obtained had such factors been investigated as well. With the present design, had any such effects existed their uncompartimentalized variability would have contributed to random variance and been incorporated into either the participant factor or the mean square error (MSE) term. This is a potential weakness of any statistical model since as the number of explicitly included factors increases so does the size and complexity of the study design. Since the statistical models and factors used in this study resulted in acceptable r^2 values and evidenced strong statistical differences amongst the observed effect levels, such possible confounding effects were not a serious limitation for this study. Nevertheless, the true sources of participant related differences in this study are unknown, and future studies are encouraged to include more highly defined anthropometric factors which may help to better identify the nature of observed participant-based effects.

5. Conclusions

The specific aims of this study were to further investigate the impacts of personal, postural, and design factors upon seat pan interface pressure. In particular, this study sought to determine whether chair design differences or postural factors accounted for greater differences in seat pan interface pressure. The final conclusion drawn from the results is that while both chair and postural treatments resulted in significant pressure distribution differences, the chair design effects yielded greater changes in interface pressure factor values than postural effects. Chair design qualities associated with the greatest pressure factor reduction were not explicitly quantified by this study, and further investigation of chair-based differences is suggested with focus upon specific engineering aspects of chair and seat pan construction. Postural treatment results indicated

a beneficial reduction in pressure factor values associated with reclined sitting. The random participant factor resulted in differences second only to the chair factor, thus further investigation of participant effects is also recommended to determine which specific attributes of participant variability resulted in such large pressure-related differences.

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